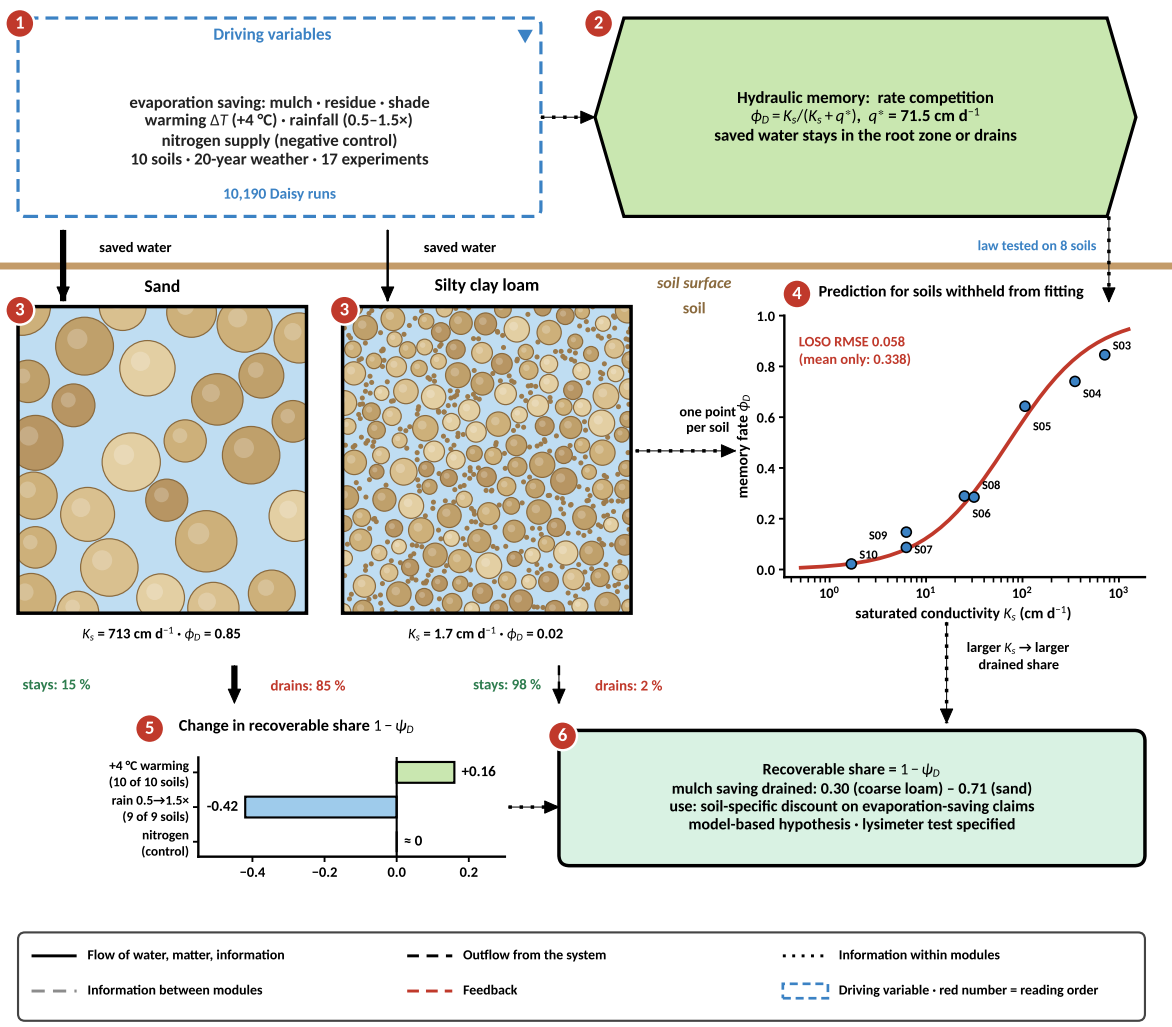
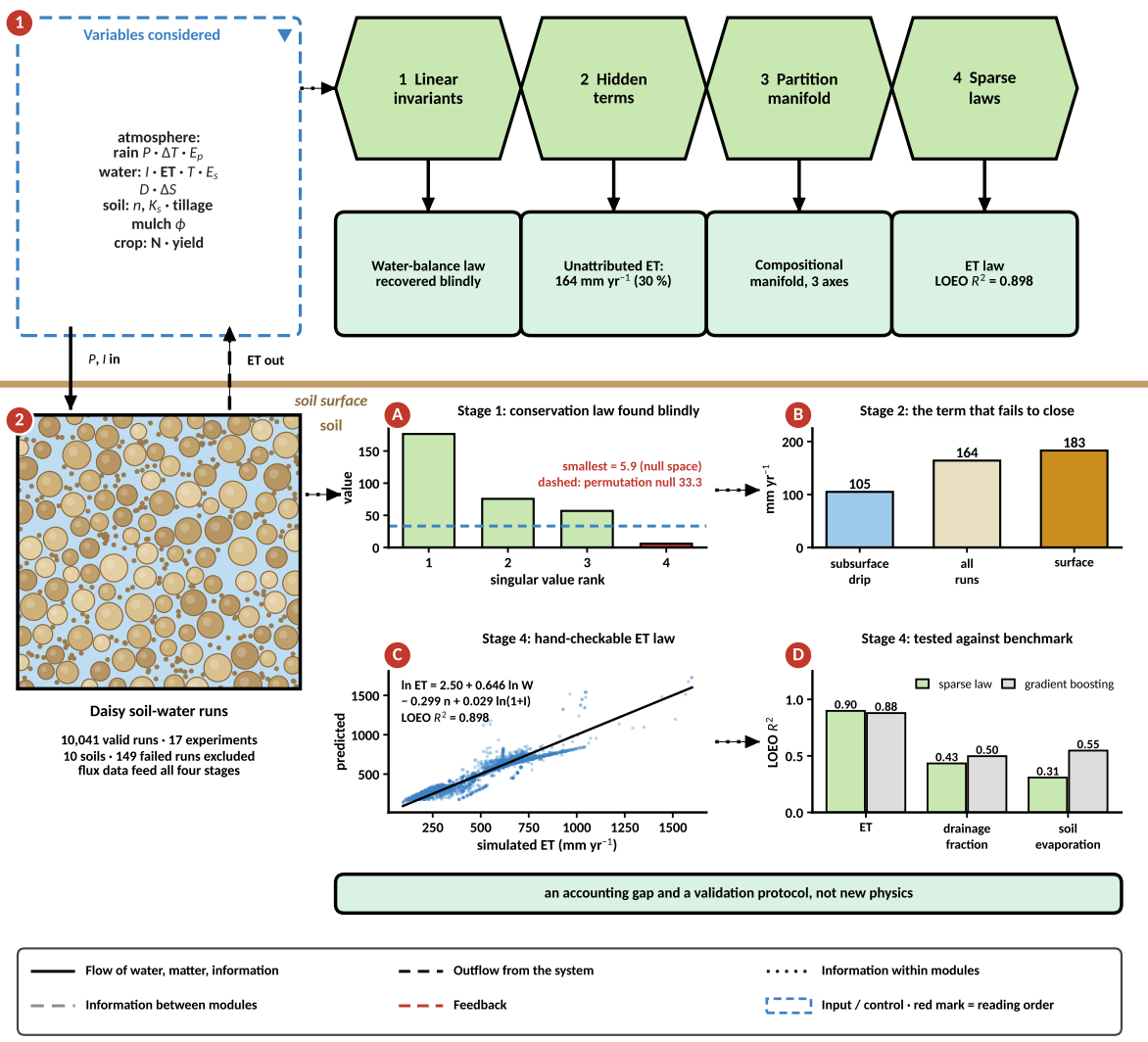
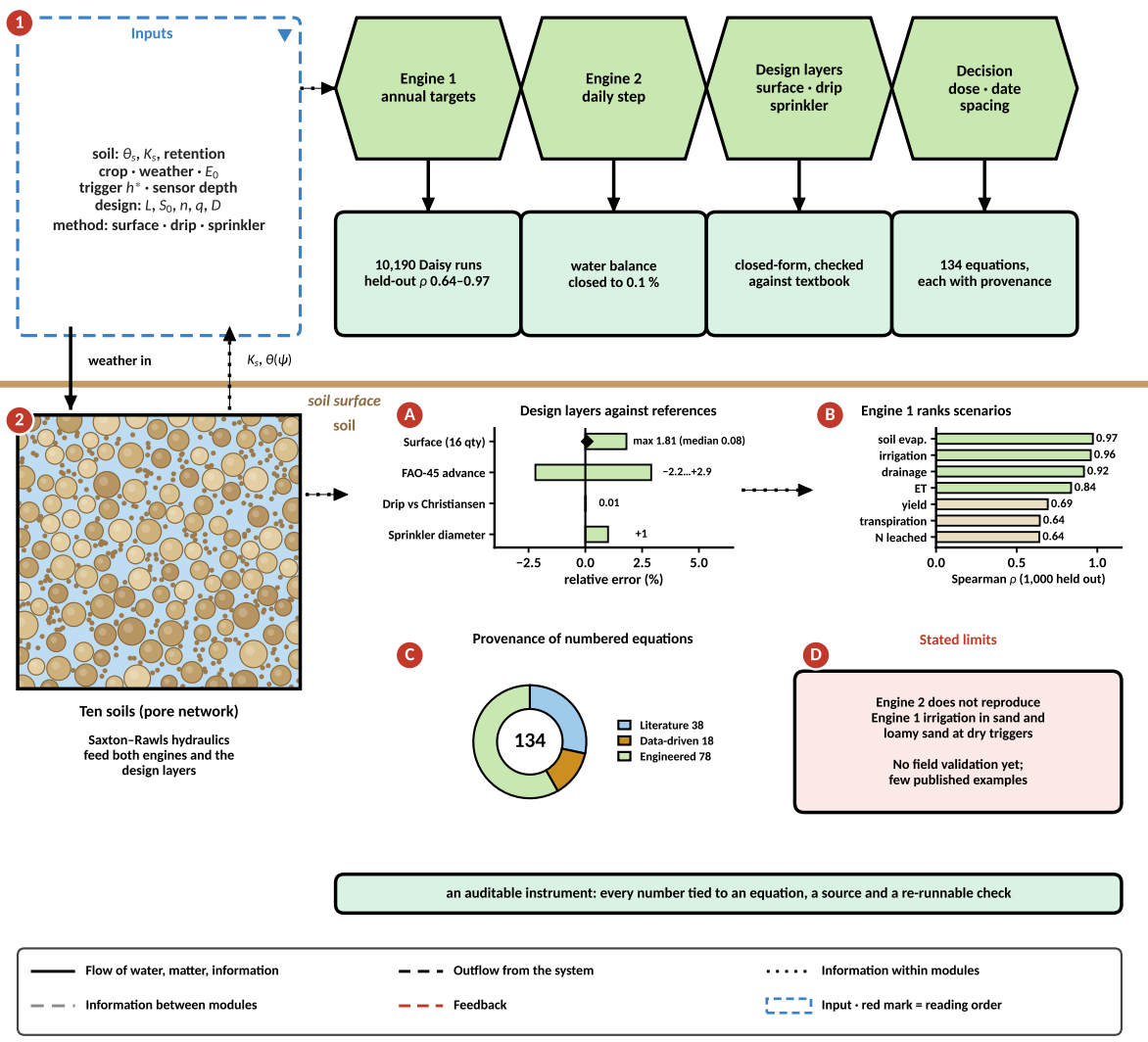

1
Driving variables
evaporation saving: mulch · residue · shade
warming ΔT (+4 °C) · rainfall (0.5–1.5×)
nitrogen supply (negative control)
10 soils · 20-year weather · 17 experiments
10,190 Daisy runs
2
Hydraulic memory: rate competition
$\phi_D = K_s/(K_s + q^*)$, q^* = 71.5 cm d^{-1}
saved water stays in the root zone or drains
law tested on 8 soils
saved water
saved water
Sand
Silty clay loam
soil surface
soil
3
3
K_s = 713 cm d^{-1} · ϕ_D = 0.85
K_s = 1.7 cm d^{-1} · ϕ_D = 0.02
one point per soil
4
Prediction for soils withheld from fitting
LOSO RMSE 0.058
(mean only: 0.338)
memory fate ϕ_D
saturated conductivity K_s (cm d^{-1})
S03
S04
S05
S08
S06
S09
S10
S07
larger K_s → larger drained share
stays: 15 %
drains: 85 %
stays: 98 %
drains: 2 %
5
Change in recoverable share 1 − ψ_D
+4 °C warming (10 of 10 soils)
rain 0.5→1.5× (9 of 9 soils)
nitrogen (control)
-0.42
+0.16
≈ 0
6
Recoverable share = 1 − ψ_D
mulch saving drained: 0.30 (coarse loam) – 0.71 (sand)
use: soil-specific discount on evaporation-saving claims
model-based hypothesis · lysimeter test specified
Flow of water, matter, information
Outflow from the system
Information within modules
Information between modules
Feedback
Driving variable · red number = reading order


1
Variables considered
atmosphere:
rain P · ΔT · E_p
water: I · ET · T · E_s
D · ΔS
soil: n, K_s · tillage
mulch ϕ
crop: N · yield
1 Linear invariants
2 Hidden terms
3 Partition manifold
4 Sparse laws
Water-balance law recovered blindly
Unattributed ET: 164 mm yr^{-1} (30 %)
Compositional manifold, 3 axes
ET law LOEO R^2 = 0.898
P, I in
ET out
2
soil surface
soil
Daisy soil-water runs
10,041 valid runs · 17 experiments
10 soils · 149 failed runs excluded
flux data feed all four stages
A
Stage 1: conservation law found blindly
smallest = 5.9 (null space)
dashed: permutation null 33.3
value
singular value rank
B
Stage 2: the term that fails to close
mm yr^{-1}
105
164
183
subsurface drip
all runs
surface
C
Stage 4: hand-checkable ET law
ln ET = 2.50 + 0.646 ln W
− 0.299 n + 0.029 ln(1+I)
LOEO R^2 = 0.898
predicted
simulated ET (mm yr^{-1})
D
Stage 4: tested against benchmark
sparse law
gradient boosting
LOEO R^2
0.90
0.88
0.43
0.50
0.31
0.55
ET
drainage fraction
soil evaporation
an accounting gap and a validation protocol, not new physics
Flow of water, matter, information
Outflow from the system
Information within modules
Information between modules
Feedback
Input / control · red mark = reading order


1
Inputs
soil: θ_s, K_s, retention
crop · weather · E_0
trigger h^* · sensor depth
design: L, S_0, n, q, D
method: surface · drip · sprinkler
Engine 1
annual targets
Engine 2
daily step
Design layers
surface · drip
sprinkler
Decision
dose · date
spacing
10,190 Daisy runs
held-out ρ 0.64–0.97
water balance
closed to 0.1 %
closed-form, checked
against textbook
134 equations,
each with provenance
weather in
K_s, $\theta(\psi)$
2
soil surface
soil
Ten soils (pore network)
Saxton–Rawls hydraulics
feed both engines and the
design layers
A
Design layers against references
Surface (16 qty)
max 1.81 (median 0.08)
FAO-45 advance
−2.2...+2.9
Drip vs Christiansen
0.01
Sprinkler diameter
+1
−2.5
0.0
2.5
5.0
relative error (%)
B
Engine 1 ranks scenarios
soil evap.
0.97
irrigation
0.96
drainage
0.92
ET
0.84
yield
0.69
transpiration
0.64
N leached
0.64
0.0
0.5
1.0
Spearman ρ (1,000 held out)
C
Provenance of numbered equations
134
Literature 38
Data-driven 18
Engineered 78
D
Stated limits
Engine 2 does not reproduce
Engine 1 irrigation in sand and
loamy sand at dry triggers
No field validation yet;
few published examples
an auditable instrument: every number tied to an equation, a source and a re-runnable check
Flow of water, matter, information
Outflow from the system
Information within modules
Information between modules
Feedback
Input · red mark = reading order